

Frontend Logic & Controllers Master Plan (Java Swing)

Assignee: Alaa (Lead Frontend Logic Developer)

Project: Hotel Booking System

UI Counterpart: Hanin (Designs the screens)

Backend Counterparts: Adham & Sherif (Build the Database & DAO)

Deadline: Phase 1 (Connecting Login & Dashboards) must be completed within 5-7 days.

Your Objective

As the lead frontend logic developer, you are the "brain" of the user interface. Hanin will build the visual shell (the buttons, text fields, and panels), and Adham will build the database logic (DAO).

Your mission is to bridge the two. You do not need to drag-and-drop UI components, and you do not need to write SQL queries. Your job is to make the buttons actually work, populate the tables with data, and validate user inputs using pure Java Swing event handling.

Phase 1: The UI Handover (Working with Hanin)

Before you write any logic, you need to access the components Hanin created.

- Hanin has been instructed to use strict naming conventions (e.g., btnLogin, txtEmail, tblRooms).
- **Your Task:** Ensure Hanin makes these components public or provides public getter methods in her JFrame classes so you can access them from your Controller classes.

Phase 2: Event Handling (Making Buttons Work)

In Java Swing, buttons don't do anything until you attach an ActionListener to them.

Example Task (Login Logic):

1. Attach a listener to Hanin's login button:
`loginScreen.btnLogin.addActionListener(e -> handleLogin());`

2. Inside the `handleLogin()` method:

- Read the inputs: `String email = loginScreen.txtEmail.getText();`
- Read the password: `String pass = new String(loginScreen.txtPassword.getPassword());`
- *Next step: Send this to Adham's DAO!*

Phase 3: The Backend Bridge (Working with Adham's DAO)

You will heavily rely on the Data Access Object (DAO) classes that Adham is building. You will instantiate these classes to fetch or save data.

How you will use Adham's code:

- **For Login:** Call `UserDAO.loginUser(email, pass)`. If it returns a valid `User` object, close the Login screen and open the `DashboardScreen`. If it returns null, show an error.
- **For Adding a Room (Admin):** Get the text from Hanin's fields, create a new `Room` object, and call `RoomDAO.addRoom(newRoom)`.
- **For Booking:** When the customer clicks "Book," get the selected room ID, create a `Booking` object, and call `BookingDAO.createBooking(newBooking)`.

Phase 4: Dynamic JTables (The Hardest Part, Made Easy)

Populating tables in Swing requires using a `DefaultTableModel`. This is a very common interview topic!

How to populate the rooms table:

1. Call Adham's method: `List<Room> availableRooms = RoomDAO.getAvailableRooms();`
2. Get the model from Hanin's table:
`DefaultTableModel model = (DefaultTableModel) dashboard.tblRooms.getModel();`
3. Clear the old data: `model.setRowCount(0);`
4. Loop through the list and add rows:

```
5. for (Room room: availableRooms) {  
    model.addRow(new Object[]{ room.getRoomNumber(), room.getType(),  
    room.getPrice() });  
}
```

Phase 5: Validation & User Feedback

You are the gatekeeper. Do not let bad data reach Adham's database!

1. **Input Validation:** Before calling any DAO method, check if the fields are empty.
 - *Example:* `if (email.isEmpty() || !email.contains("@")) { ... }`
2. **Swing Dialogs (Alerts):** If the user does something wrong or if a booking is successful, give them visual feedback using `JOptionPane`.
 - *Success:* `JOptionPane.showMessageDialog(null, "Booking Confirmed! ""Success," JOptionPane.INFORMATION_MESSAGE);`
 - *Error:* `JOptionPane.showMessageDialog(null, "Invalid Email or Password.", "Error", JOptionPane.ERROR_MESSAGE);`

Alaa's Golden Rules:

1. **Try-Catch Everywhere:** Adham will throw custom exceptions (like `DatabaseException` or `RoomNotAvailableException`). You must wrap your DAO calls in try-catch blocks and show the error message in a `JOptionPane` so the application doesn't crash.
2. **Don't Freeze the UI:** If fetching data from the database takes a second, the Swing screen might freeze. For bonus marks, research how to use `SwingWorker` to run database tasks in the background!

Next Steps for You:

Communicate with Hanin today to agree on the exact names of the text fields and buttons she is creating. Prepare your controller classes so they are ready to receive Hanin's UI and Adham's DAO. You are the glue that holds this app together!